Change Log: Synthetic Data and Check Scripts, v1 → v2

Gallup BI Agent project · October 2026 · Issues #1–25 from the team review (Notion “Version1 (Selvi part's)”), plus #26

Issues **26**	High **3**	Medium **14**	Low **9**	Status: all fixes applied and verified on the v2 data

#	Problem	Impact	Severity	Fix applied	Evidence in v2	Task
generate_synthetic_data.py (generator)						
1	X7 signal too weak: who answers the survey depended only on whether they later left, not on engagement.	The agent could not detect that survey numbers look too positive; true and respondent ORs were both 0.36.	High	• Two new settings: engagement slope 1.5 for leavers, 0.3 for stayers. • Each person's response probability now depends on leaving and on hidden engagement, calibrated so group rates stay at 60% and 86%. • ground_truth.json records the X7 design and both values.	Mean-score OR 0.36 true vs 0.52 among respondents (2M).	P2-09 Done
2	Large performance gap between departments (Sales 4.6, Logistics 2.3).	Data looked unrealistic.	Medium	• Kept on purpose: weakening it makes the Simpson's paradox disappear (tested). • Spec now explains X6 as departments rating with different leniency, a real HR problem.	Gap unchanged by design; Simpson's paradox present (pooled −0.12, within +0.17).	P2-10 Done
3	Every department had the same safety incident rate.	Office departments had workplace injuries.	Medium	• Safety removed. Outcomes are now turnover, absenteeism and performance, which every HR system holds for every employee.	team_outcomes.csv holds only team id, department and size.	P2-10 Done
4	Every team had a CSAT score.	Finance and HR had customer satisfaction.	Medium	• CSAT removed (with #3). The team file is kept for small-team suppression (X8) and clustered errors.	Region column also removed.	P2-10 Done
5	All 12 items had identical score distributions.	Did not look like real Q12 data.	Low	• Item-specific offsets (Q01, Q08, Q09 easier; Q04, Q06, Q10, Q11 harder). • One shared set of cut points for all 12 items.	Agree share now ranges from 41% (Q10) to 72% (Q08).	P2-10 Done
6	Manager quality affected only Q07.	Unlike research where managers shape overall engagement.	Low	• Kept so trap X3 stays measurable; the spec states the simplification and the agent always checks team-level effects for absence.	Documented in spec B4.	P2-10 Done
7	Department productivity effects were redrawn every wave.	Department traits changed each year.	Low	• Resolved by removing productivity and CSAT (#3, #4).	Team facts are fixed across waves.	P2-10 Done

#	Problem	Impact	Severity	Fix applied	Evidence in v2	Task
validate_synthetic_data.py (validator)						
8	N3 (Q08 unrelated to all outcomes) was never checked.	No proof the answer key was right.	Medium	• Three N3 checks (turnover, absence, performance), each confirming Q08 is associated alone but has no independent effect. • N1 now uses the full 12-item model; N2 retargeted to Q10 across all three outcomes.	All null checks pass on 5K and 2M.	P2-14 Done
9	X3 (hidden manager) was never checked.	No proof the trap worked.	Medium	• Four checks: Q07 linked to absence with department controlled; the link shrinks within teams; absence clusters by team; controlling hidden manager quality removes Q07.	2M: Q07 coefficient −0.146 → 0.000 within teams; team ICC 0.10.	P2-14 Done
10	2M package validator read three files, but the package shipped one merged file per wave.	Nobody could rerun validation on 2M.	Medium	• 5K and 2M now share one file layout (three agent_data files) and one validator.	Validator runs on every dataset folder.	P2-14 Done
11	Hard-coded data path and small-team count of 20.	Code edits needed for each dataset.	Low	• Folder passed on the command line; small teams checked as a share (about 5%). • Expected values read from ground_truth.json.	Same script validates 9 datasets.	P2-14 Done
bias_validation.py → health_check.py (health check)						
12	Could not read the 2M merged format.	Failed on the 2M package.	Medium	• Fixed by the unified file layout (#10).	Runs on 2M in about 40 seconds.	P2-13 Done
13	“Outcomes measured after the survey” always printed PASS without checking.	Wrong time order in real data would go unnoticed.	High	• Date columns added (survey date and each outcome's window end). Row-by-row comparison; FAIL blocks that outcome only; a missing date column is a WARN, never a PASS.	dirty_A, B, C catch every planted date error.	P2-13 Done
14	The leaver warning text was fixed in the code.	Printed regardless of the data.	Low	• Message chosen from the evidence in four cases (similar rates; leavers lower; leavers lower and non-respondents worse off; leavers higher).	2M shows the strong case: non-respondent absence SMD −0.20.	P2-13 Done
15	CSV output only, nothing for the agent.	The agent could not apply warnings automatically.	Medium	• New health_report.json with code, status, metrics, evidence strength, affected outcomes and actions.	2M: 45 checks, 33 PASS, 12 WARN, 0 FAIL.	P2-13 Done

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16	FAIL had never been triggered.	Unknown whether real errors would be stopped.	Medium	• Three dirty test sets with error logs: A (one of each), B (many, overlapping, missing date column), C (boundary values). • The validator compares each error log with the health report.	All planted errors caught with exact counts; no false FAIL on clean data.	P2-12 Done
2M data package and answer key						
17	README still said 400 teams and 20 small teams.	Did not match the 80,000 teams.	Low	• README written by the generator from the real counts.	2M README: 80,000 teams, 4,000 small.	P2-15 Done
18	Answer key had no hidden engagement scores.	Could not compare true and respondent effects on 2M.	Low	• Realized values (all employees and respondents) computed during generation and stored in ground_truth.json; no 2M full-data file; seed and version recorded.	ground_truth.json holds 19 realized values per group.	P2-09 Done
Gallup_Agent_Matrix_A_B.pdf (spec)						
19	Effect values were draft numbers (for example T1 0.75 vs 0.70).	Grading against the PDF would be wrong.	High	• B2 built from ground_truth.json with planted and realized values; T5–T7 removed; T5 (Q05 → absence) added.	T1 realized OR 0.62 (respondents, 2M).	P2-15 Done
20	Only the 5,000-employee version was described.	The 2M version was missing.	Medium	• B1 lists 5K and 2M side by side; small teams described as a share.	Spec v2 section B1.	P2-15 Done
21	Turnover listed survival analysis, but there are no leave dates.	Method could not be run.	Medium	• Logistic regression on leaving within 12 months; survival analysis moved to future work.	Spec v2 section A2.	P2-15 Done
22	B1 said outcomes cover 12 months after each wave, but absence covers 6.	Contradicted A2.	Low	• Each outcome's own window stated: turnover 12 months, absence 6 months, performance next review.	Spec v2 sections A2 and B1.	P2-15 Done
23	No effect thresholds, applicability rules or test datasets.	Rules were incomplete.	Medium	• New sections: effect thresholds (proposed: ratio outside 0.91–1.10, |β| of 0.05 or more), applicability rules, test dataset list.	Spec v2 sections A4, A5, B5. Thresholds await team sign-off.	P2-15, P2-17, P2-19 Done
Data_Validation_Bias_Checks.docx (health check guide)						
24	Said the checks caught 5 of 8 traps; the correct count is 4.	A visible error for judges.	Low	• States 4 of 8 (X1, X6, X7, X8); the other 4 (X2–X5) are handled in modeling.	Matches deck Table 11.	P2-08 Done

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25	“Turnover effects understated” was not true under the v1 data design.	Incorrect claim.	Medium	• v2 data makes it true (#1); Step 3 explains the full chain; the message now comes from the evidence (#14).	Respondent OR 0.52 vs true 0.36.	P2-08, P2-09 Done
Decision added during review						
26	Performance analysis did not control for department.	Department rating leniency (X6) distorts performance results.	Medium	• Performance controls are now tenure, department and prior rating, in code, spec and deck Table 9.	stats_tools.performance_model.	P2-15 Done

Verification: validate_synthetic_data.py passes every check on all nine datasets (5K 31/31, 2M 31/31, null 19/19, weak 31/31, seed2 31/31, seed3 31/31, dirty_A 10/10, dirty_B 10/10, dirty_C 5/5). The 5K practice set uses seed 2: at 5,000 employees, 11 of 20 seeds pass every check, and the misses are chance results at small sample size, which is why formal grading uses 2M (seed 1). statsmodels could not be installed in the build environment, so stats_tools.py implements the models with numpy and scipy; results were checked against scikit-learn.